

AERODYNAMICS

DESCRIPTION: Teams will build on site paper airplanes using provided materials and test airplanes based on flight duration and accuracy through a target. This is an On-Site Build event.

COMPETITORS BRING:

- pencils/writing utensils;
- Scissors (one pair per team)
- Ruler (one per team)

NUMBER OF PARTICIPANTS: 2

IMPOUND: NO (ALL TEAMS BUILD ON-SITE)

THE COMPETITION:

1. Teams will construct (on-site) paper airplanes designed to meet the goals outlined.
2. All teams will construct at the same time to avoid an unfair advantage to those that test first.
3. All teams will return at a later point in the day (as outlined by the schedule) to test their planes.
4. It is suggested that Event Supervisors set up at least two separate areas to test - one for Part 1 and one for Part 2. This will ensure that teams will have adequate time to test in the testing block.

BUILD (20 minutes):

1. Each team member receives 2 pieces of copy paper, 2 paper clips, and 5 cm of masking tape and has 20 minutes to build their airplanes. Type/size of paper clip and width of tape will be consistent among all teams.
2. Each team member constructs his or her own airplane/s - using at least one sheet of paper. But not more than two, which will bear a resemblance to a winged object. No designs like a helicopter are permitted.
3. Teams will write their team number on each plane.
4. One team member's plane/s will be used in Part 1 with the goal being to have the plane in the air the longest. These planes should be labeled "Part 1".
5. The other team member's plane/s will be used in Part 2 with the goals of accuracy to hitting targets. These planes should be labeled "Part 2".
6. Team members should decide whose plane will be used for each part of the competition prior to testing time.
7. The event supervisor should notify teams of where to leave their planes for use during later testing. This location should be monitored so planes are not touched prior to testing.
8. It is suggested that the event supervisor/judge does not touch student planes and instead gives students instructions on where to place them.
9. Teams must clean up after themselves and may be placed in Tier 2 if the construction area is not clean.
10. Teams should leave their design log with the event supervisor during this time.

TESTING Part 1 (Duration of Flight)

1. The goal of this part is for the airplane to stay in the air the longest amount of time.
2. Each plane is launched by hand by its maker from a predetermined starting point, which is the same for all teams. This starting point will be designated by a 1 meter square taped on the floor.
3. Teams should be prepared to launch within 60 seconds of being called upon.
4. Teams may choose to launch one plane at a time (two launches) or launch both planes at once in a "piggyback fashion" one launch
5. If two planes are used at once (in piggyback fashion), the scored time will be for the plane which is aloft for the longer duration.

6. There is no time provided for a test run - a maximum of two launches are allowed to get a maximum of two scores for aloft time. The greater flight time will count as the score for Part 1.
7. Launching team member must give a warning by calling out "3, 2, 1, Launch" or something similar at the time of launch. Failure to do so may result in the loss of the attempt. If this happens, the Event Supervisor should provide a reminder before the next attempt.
8. The time aloft will cease when:
 - a. A plane becomes lodged in a permanent fixture (light, seat, etc)
 - b. A plane touches the floor.
 - c. The plane ceases moving in air or hits a hanging obstacle.

TESTING Part 2 (Accuracy of Flight)

1. The purpose of this part is for the airplane to fly accurately through targets.
2. The planes used in this part must be different from the planes used in Part 1 and should be made by the other team member than in Part 1.
3. Each plane is launched by hand by its maker from a predetermined starting point, which is the same for all teams. This starting point will be designated by a 1 meter square taped on the floor.
4. Teams should be prepared to launch within 60 seconds of being called upon.
5. Teams may choose to launch one plane at a time (two launches) or launch both planes at once in a "piggyback fashion" (one launch)
6. If two planes are used at once (ie. in a piggyback fashion), the accuracy score will be for the plane that goes through the most targets.
7. There is no time provided for a test run - a maximum of two launches are allowed to get maximum score.
8. Two to three targets made from hoops will be erected. Points will be awarded for each hoop through which the airplane passes during its flight.
 - a. The hoops may be of any size (from hula hoops to ones constructed from flexible PVC) but must be stationary and not held by a person. For example, hoops can be held up using purchased hula hoop holders or holders made from a 2-liter bottle with rocks or sand that is attached to the hoop with PVC.
 - b. The distance from the launch area to the initial hoop must be no greater than 3 meters and the distance to the final hoop should be no greater than 10 meters.

DESIGN LOG:

Prior to the competition, teams should prepare a design log. The design log will be submitted during the Construction Block and should include:

- a. Graphs or data charts compiled from flight practice for each part of the competition, graph title, labeled x- and y- axes and using appropriate units for usage of their practice flights.
- b. The log should be clearly labeled with the team number.

Teams who do not submit a design log will be placed in Tier 2.

RELATED FLORIDA STANDARDS:

- **SC.5.P.13.2** Investigate and describe that the greater the force applied to it, the greater the change in motion of a given object.
- **SC.4.P.12.2** Investigate and describe that the speed of an object is determined by its travels in a unit of time and that objects can move at different speeds.
- **SC.5.P.13.3** Investigate and describe that the more mass an object has, the less effect a given force will have on the object's motion.
- **SC.5.P.13.4** Investigate and explain that when a force is applied to an object but it does not move, it is because another opposing force is being applied by something in the environment so that the forces are balanced.

SCORING:

1. High score wins.
2. Scoring as follows:
 - a. Score for Part 1 = Longest flight time in seconds X 10
 - b. Score for Part 2 = 50 points for each target through which the plane passes (up to 3)
 - c. Tiebreaker = Quality of design log

Team# _____

School Name: _____

Aerodynamics Scoring Sheet

BUILD:

		TIER
1. Outside assistance violation?	Yes= Tier 2	
2. Planes made only of allowable materials?	No= Participation points only	
3. Planes have the team number and Part 1 or 2 on them?	No= Tier 2	
4.. Building area cleaned up as instructed?	No = Tier 2	
5. Both team members attend the build?	No= Participation points only	
6. Team impounded a data chart/log?	No = Tier 2	

TEST:

Part 1: Duration of flight Launch 1: _____ seconds Launch 2: _____ seconds Score will be 10 times the greater of the two launches.	
Part 2: Points for flying through targets Launch 1: _____ targets Launch 2: _____ targets Score will be given only for launch with the most accuracy. Teams will earn 50 points per target up to 150 points total.	
Design Log: Used for tiebreak only.	
Total Testing Score=	

TESTING VIOLATIONS:

1. Outside assistance violation?	Yes= Tier 2	
2. Design alteration violation during testing phase?	Y= Participation points only	
3. Launch made outside the designated point?	Yes= Tier 2	

Total Testing Score: _____ Tier: _____ Final Rank: _____

PLEASE NOTE THAT THIS RUBRIC IS A SAMPLE AND GUIDE- IN CASES OF CONFLICT WITH THE RULES, THE
RULES TAKE PRECEDENCE OVER THE RUBRIC.